

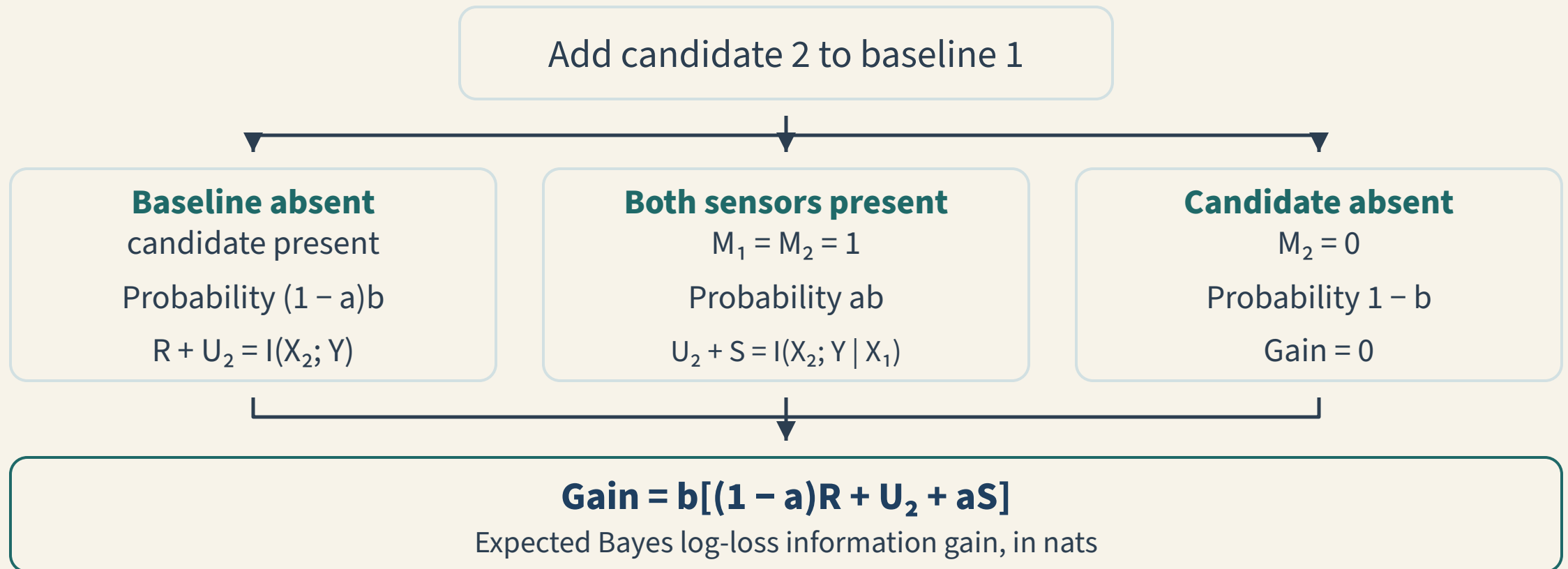
When a second sensor becomes useful

R: redundancy · U_2 : unique candidate term · S: synergy (signed MGW atoms)

$P(X_1, X_2, Y)$ fixed; $M \perp (X_1, X_2, Y)$. The sensors may be dependent.

$M_i = 1$ means sensor i is available; $a = P(M_1 = 1)$, $b = P(M_2 = 1)$.

The product probabilities below also require M_1 and M_2 to be independent.



Correlated masks: $\text{Gain} = p_{01}(R + U_2) + p_{11}(U_2 + S)$; $p_{ij} = P(M_1 = i, M_2 = j)$.

Masks still independent of data. Natural occlusion needs conditional-law review.

This identity does not prove benefit for a learned prediction or control policy.